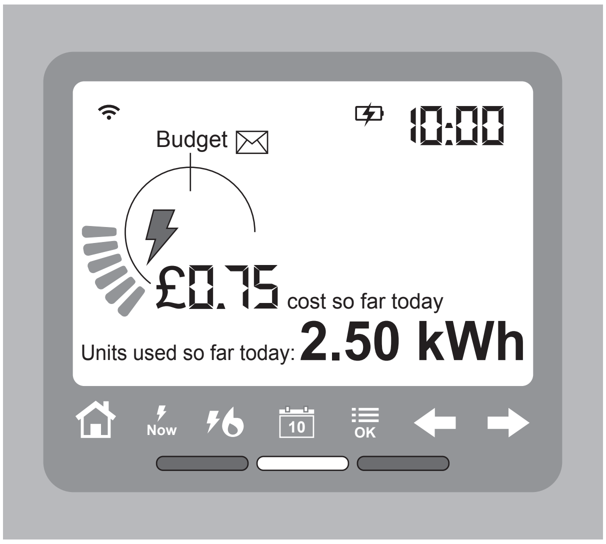
Examiner only

5. Smart meters are fitted into some houses to measure how much electricity is used. Householders can monitor and try to reduce their electricity use.
The diagram of the smart meter below shows the units used in **10 hours** from midnight to 10.00 am.



(a) (i) Use the equation:

$$\text{mean power (kW)} = \frac{\text{units used (kWh)}}{\text{time (h)}}$$

and information from the diagram to calculate the mean power of the appliances used in the 10 hours. [2]

mean power = ... kW

 (3420U10-1)

Examiner only

(ii) I. State the cost of the electricity used in 10 hours in pence (p). [1]

cost = ... p

II. Use the equation:

$$\text{cost of 1 unit (p)} = \frac{\text{cost (p)}}{\text{units used (kWh)}}$$

to calculate the cost of 1 unit of electricity. [2]

cost of 1 unit = ... p

(b) Give **one** reason, **other than** to save money, why householders should be encouraged to reduce their electricity use. [1]

....

....

(c) Describe the function of an **earth wire**, a **fuse** and a **residual current circuit breaker (rccb)**.

Include in your answer:
- what they are each used to protect
- the fault that causes each of them to work. [6 QER]

12